

TITLE: Model Comparison for House Price Prediction Using Machine Learning

Name: Saksham Chaudhary

Course: BCA AIML

Subject: Applied Machine Learning

Assignment: Task 2 – Model Comparison

1. Objective

The objective of this assignment is to compare different machine learning regression models and identify the best-performing model for predicting house prices. The California Housing Dataset was used for training and evaluation.

2. Dataset Description

The California Housing Dataset contains information related to housing prices in California districts. The dataset includes several features such as median income, house age, average rooms, average bedrooms, population, occupancy, latitude, and longitude.

Target Variable:

- HousePrice

Features:

- MedInc
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup
- Latitude
- Longitude

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	HousePrice
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

3. Data Preprocessing

The following preprocessing steps were performed:

- Dataset loaded successfully.
- Data structure examined using `shape()`, `info()`, and `describe()`.
- Missing values checked using `isnull().sum()`.
- Features and target variable separated for model training.

4. Train-Test Split

The dataset was divided into training and testing sets using Train-Test Split.

- Training Data: 80%
- Testing Data: 20%
- Random State: 42

The training dataset was used for model learning, while the testing dataset was used to evaluate model performance on unseen data.

5. Machine Learning Models Used

The following regression models were compared:

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

Each model was trained using the training dataset and evaluated using performance metrics.

6. Evaluation Metrics

The following evaluation metrics were used:

Mean Absolute Error (MAE):

Measures the average absolute difference between actual and predicted values.

Root Mean Squared Error (RMSE):

Measures prediction error while giving higher weight to larger errors.

R² Score:

Measures how well the model explains the variation in the target variable.

7. Results

The performance comparison of the models is shown below:

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

Based on the evaluation metrics, the Decision Tree Regressor achieved the lowest RMSE and highest R² Score.

Best Model:

Decision Tree Regressor

The trained model was successfully saved as:

best_model.joblib

	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

Model Saved Successfully

8. Visualization

The following visualization was generated:

- Actual vs Predicted House Prices Plot

This visualization helped evaluate the prediction performance of the trained model.



9. Conclusion

Different regression algorithms were trained and compared for house price prediction using the California Housing Dataset. After evaluating all models, the Decision Tree Regressor achieved the best performance with the lowest RMSE and highest R^2 Score.

The model demonstrated better predictive capability compared to Linear Regression and Ridge Regression. The trained model was successfully saved for future use and deployment.

Hence, the Decision Tree Regressor was selected as the final model for house price prediction.